

THE BRIEF

AUDIENCE

Funders & employer partners;
Reach / NCAD program design

FRAMING

Program doctrine — design
backward from demand

THE REFRAME

Don't assemble from a catalog —
bet opinionatedly on frontier fields

THE ASK

Sequence first builds with employer
partners in three frontier fields

KEY SIGNALS

93%
RA completers
employed • NIST

~\$77K
Avg. starting wage •
NIST

2.1M
Mfg jobs unfilled by
2030 • NAM

~\$164B
U.S. drone industry by
2030

THREADS

- Advanced Mfg
- Robotics
- Drone Sciences
- Applied AI
- Earn-and-Learn
- Employer Pull

Figures are from industry / labor reports and
flagged for verification against primary sources.
Pick 1–2 hero stats per field before presenting.

Design backward from the job, not forward from
the catalog.

Apprenticeship-degree programs should be designed toward where job demand is actually going — opinionated bets on fields with large unmet demand, real employer pull, and work that suits a work-embedded degree. Three frontier fields score high on all three: Advanced Manufacturing Operations, Drone Sciences, and AI Technology in Business Operations.

- ▶ **The field outruns the catalog.** Robotics, drones, and applied AI change faster than course revisions — work-embedded learning stays current by design.
- ▶ **Employers are the co-designers.** Frontier demand means employers are desperate for talent now and will sponsor, mentor, and hire — the pull that makes apprenticeship work.
- ▶ **Outcomes are already proven.** ~93% of registered-apprenticeship completers gain employment at ~\$77K average starting wage (NIST); debt-free earn-and-learn widens the funnel.

01 ADVANCED MANUFACTURING

- **2.1M** mfg jobs could go unfilled by 2030; ~94% report a skills gap (Deloitte/NAM).
- **97,500** advanced-mfg apprentices in 2025, +20% over five years.
- **Shape:** software-driven lines, robotics/cobots, observability.

03 AI IN BUSINESS OPS

- **~90% / 9%** of orgs use AI but are "AI-mature" — a huge operationalization gap (Gloat).
- BCG: **50–55% of U.S. jobs reshaped** by AI in 2–3 years.
- **Shape:** applying AI inside operations, not building frontier models.

02 DRONE SCIENCES

- **~\$164B** U.S. drone industry by 2030 (~13% CAGR); ~650K drone-linked jobs (PwC).
- **Shape:** manufacture, software stack, field deployment, sector applications.
- Anchored by the FAA Part 107 remote-pilot credential.

04 THE DESIGN PRINCIPLE

Pick fields by **demand signal × employer pull × apprenticeship fit**, then design backward from the job: start with the role an employer is hiring for, work out the competencies, build the degree around the day-to-day work. **Say no** to fields that don't clear all three bars.

OPINION IS
THE
PRODUCT

A program that tries to serve everyone serves no employer well. Point new program design at frontier fields where **earn-and-learn pays off fastest**, map each to specific NCAD member employers who'd co-sponsor, and sequence the first builds with them. **Opinion is the point** — the curriculum lives in the work because the work is where the field stays current.

Sources: NAM, "2.1 Million Manufacturing Jobs Could Go Unfilled by 2030" • Deloitte, manufacturing skills-gap study • Apprenticeship.gov, "Advanced Manufacturing" (97,500; +20%) • NIST, regional registered-apprenticeship (93% / \$77K) • Pilot Institute, drone statistics (\$163.6B) • BCG, "AI Will Reshape More Jobs Than It Replaces" • PwC, "AI Jobs Barometer" • Gloat, "AI Skills Demand" — *draft narrative; figures pending verification.*